

DECK 01 · WEEK 1 · MORNING 1

Time Value of Money & DCF

The financial-math foundation every deck that follows will draw from

CFA Level I · Vol 1 Learning Modules 1–2

Roadmap — Time Value of Money & DCF Foundations

6 sections · ~2 hours · Built bottom-up from CFA readings

01

What is an interest rate?

3 interpretations · decomposing r · risk premia · CFA Example 1 five bonds · nominal vs real (Fisher)

02

Measuring return

HPR · arithmetic / geometric / harmonic · trimmed & winsorized · money- vs time-weighted

03

Annualizing returns

periodic $\rightarrow$ annual · EAR · continuous compounding · gross / net / after-tax / real / levered

04

Future value

single CF · compounding frequency · FV of an annuity (retirement savings)

05

Present value

single CF · annuity · perpetuity · India & Greece bonds · Shipline DDM · implied r and g

06

NPV, IRR & applications

NPV decision rule · when IRR breaks · mortgage amortization · TVM archetype map

Learning objectives

By the end of this deck you should be able to:

1. Read an interest rate three ways — required return, discount rate, opportunity cost — and back out each risk premium from a set of traded bonds.
2. Choose the correct mean (arithmetic, geometric, harmonic, trimmed / winsorized) for a given return-aggregation problem and justify the choice.
3. Distinguish money-weighted from time-weighted returns and explain why the latter is the standard for evaluating managers.
4. Convert any periodic return into an annualized return, an EAR, or a continuously compounded rate, and move a cash flow anywhere on a timeline.
5. Price a bond, value a stock with a constant- or two-stage DDM, and run both backwards to recover the market-implied return and growth rate.

SECTION 01

What is an interest rate?

An interest rate is simultaneously a required return, a discount rate, and an opportunity cost — three views of the same number

If USD 9,500 today and USD 10,000 in one year are equivalent in value, then $r = 500 / 9,500 = 5.26\%$.

$$r = (FV - PV) / PV \rightarrow (10,000 - 9,500) / 9,500 = 5.26\%$$

1

Required rate of return

The MINIMUM rate an investor must receive to accept an investment. Below this rate, they walk away.

2

Discount rate

The rate at which USD 10,000 in one year is equivalent to USD 9,500 today — the same operation run backwards.

3

Opportunity cost

The value forgone by choosing a course of action. Spending the USD 9,500 today forgoes the 5.26% earned by saving it.

Every market interest rate decomposes into a real risk-free rate plus four risk premia

The CFA states this decomposition additively. The ranges below are illustrative teaching anchors, not the curriculum.

$$r = r_{\text{real-rf}} + \text{Inflation} + \text{Default} + \text{Liquidity} + \text{Maturity}$$

Component	Illustrative	What it compensates for
Real risk-free rate	0 – 2%	Time preference, current vs future consumption
Inflation premium	2 – 3%	Expected inflation over the LIFE of the bond
Default risk premium	0 – 5%+	Risk the borrower fails to pay as contracted
Liquidity premium	0 – 1.5%	Loss vs fair value if you must sell quickly
Maturity premium	0.5 – 2%	Rising price sensitivity as maturity extends

TWO THINGS THE TABLE HIDES

US T-bills bear NO liquidity premium — size can be traded without moving the price. And every premium, including the real risk-free rate, moves continuously: the decomposition is a lens, never a fixed set of numbers.

Source: CFA Institute, Vol 1, Learning Module 1, "Determinants of Interest Rates", pp. 6–7.

CFA LM 1 Example 1 lets us back out each premium by comparing five bonds that differ on one dimension at a time

All five promise a single payment at maturity. Inflation, liquidity and default premia are constant across horizons.

Inv.	Maturity	Liquidity	Default	r (%)
1	2y	High	Low	2.0
2	2y	Low	Low	2.5
3	7y	Low	Low	r_3 ?
4	8y	High	Low	4.0
5	8y	Low	High	6.5

THE TECHNIQUE, NOT THE ANSWER

Hold everything constant but one feature, take the difference, and the market has priced that feature for you.

This is the whole of relative value: comparables exist so that a single variable can be isolated.

SOLUTION – READ THE PREMIA OFF THE MARKET

Compare bonds that differ on exactly one feature to isolate each premium.

1. Liquidity premium: $\text{Inv 2} - \text{Inv 1} = 2.5\% - 2.0\% = 0.50\%$ (same maturity, same default)
2. Default premium: $(\text{Inv 5} - \text{Inv 4}) - \text{LP} = (6.5\% - 4.0\%) - 0.50\% = 2.00\%$
3. Lower bound on r_3 : Inv 2 has the same liquidity and default but is shorter $\Rightarrow r_3 > 2.50\%$
4. Upper bound on r_3 : Inv 4 re-quoted with low liquidity = $4.0\% + 0.50\% = 4.50\% \Rightarrow r_3 < 4.50\%$

$$\Rightarrow 2.50\% < r_3 < 4.50\%$$

The nominal risk-free rate is the real rate compounded with expected inflation — the additive version is only an approximation

Real 2% + 3% inflation: exact 5.06% vs additive 5.00%, ignorable. At 50% inflation: 53.0% vs 52.0%, not ignorable.

$$(1 + r_{\text{nominal-rf}}) = (1 + r_{\text{real-rf}}) \times (1 + \pi) \approx r_{\text{real-rf}} + \pi$$

where: π = expected inflation over the horizon of the instrument. The dropped term is the cross-product $r_{\text{real}} \times \pi$.

UNITED STATES

The 90-day T-bill is the standard nominal risk-free proxy for USD analysis. A 90-day bill quoted at 3% is the ANNUALIZED rate — not the return earned over the 90 days.

FRANCE

BTFs — Bons du Trésor à taux fixe et à intérêts précomptés — are negotiable fixed-rate discount Treasury bills with maturities up to one year. The eurozone short-horizon proxy.

JAPAN

The Japanese government issues short-term Treasury bills at 6- and 12-month maturities: the JPY short-dated risk-free proxy.

Source: CFA Institute, Vol 1, Learning Module 1, "Determinants of Interest Rates", pp. 6–7.

SECTION 02

Measuring return

A holding-period return captures both income yield and capital gain over any single period

HPR is the single-period building block — every multi-period measure that follows either averages or compounds HPRs.

$$R = (P_1 - P_0 + I_1) / P_0 = \text{capital-gain yield} + \text{income yield}$$

where: P_0 = beginning price; P_1 = ending price; I_1 = income (dividend, coupon) received at period end.

CFA LM 1 EXAMPLE 2 — SINGLE-QUARTER EQUITY POSITION

Investor buys 100 shares at USD 34.50, receives a USD 51.55 dividend, sells at USD 30.50 one quarter later.

1. Cost basis: $100 \times 34.50 = \text{USD } 3,450$
2. End-of-period value + income: $(100 \times 30.50) + 51.55 = \text{USD } 3,101.55$
3. $\text{HPR} = (3,050 - 3,450 + 51.55) / 3,450 = -10.1\%$
4. Split it: income yield $51.55 / 3,450 = +1.49\%$; capital loss $-400 / 3,450 = -11.59\%$

$$\text{HPR} = -10.1\% = +1.49\% \text{ income} - 11.59\% \text{ capital}$$

¹ Multi-period HPRs link multiplicatively: $R = (1+R_1)(1+R_2)(1+R_3) - 1$. CFA Example 3: +14%, -10%, -2% compounds to +0.55%, not the +2% a naive sum gives.

Source: CFA Institute, Vol 1, Learning Module 1, §3 "Rates of Return", Eq. 1 and Example 2, pp. 9–11.

Arithmetic mean is the right estimate of NEXT period's return — and a biased overstatement of what capital actually did

CFA Exhibit 2 — three annual returns of -50%, +35%, +27% on one euro of capital.

$$\bar{R}_i = (R_{i1} + \dots + R_{iT}) / T = (-50 + 35 + 27) / 3 = +4.00\%$$

Year	Actual return	Actual year-end amount	At AM of +4%	At GM of -5%
Year 0	—	EUR 1.0000	1.0000	1.0000
Year 1	-50%	0.5000	1.0400	0.9500
Year 2	+35%	0.6750	1.0816	0.9025
Year 3	+27%	0.8573	1.1249	0.8574
Verdict	AM = +4.00%	truth = 0.8573	+0.2676 too high	GM = -5.00% ✓

WHY THE ARITHMETIC MEAN IS BIASED UP

The AM assumes the same amount is invested at the start of every period. Capital does not work that way: the +35% in Year 2 is earned on EUR 0.50, not on the original EUR 1.00. The bias is zero only when every period return is identical, and it is worst when gains and losses alternate.

Source: CFA Institute, Vol 1, Learning Module 1, §3 "Arithmetic or Mean Return" and Exhibit 2, pp. 9–10.

Geometric mean return captures the actual compound growth of one unit of capital – always $\leq$ arithmetic, equal only when every return is identical

The gap between AM and GM widens with dispersion. It is a volatility tax, and it is paid in real money.

$$\bar{R}_G = [(1 + R_1)(1 + R_2) \dots (1 + R_T)]^{(1/T)} - 1$$

where: Compound the GROSS factors (1+R), never the raw returns. The product must stay non-negative — a -100% period ends the series.

SAME SERIES AS THE PREVIOUS SLIDE: -50%, +35%, +27%

Compute the geometric mean and check that it reproduces the actual EUR 0.8573 ending balance.

1. Gross factors: $(1 - 0.50) \times (1 + 0.35) \times (1 + 0.27) = 0.50 \times 1.35 \times 1.27 = 0.85725$
2. Cube root: $0.85725^{(1/3)} = 0.9500 \rightarrow \bar{R}_G = -5.00\%$
3. Check: $1.0000 \times (1 - 0.05)^3 = 0.8574$ ✓ matches the actual balance

GM -5.00% vs AM +4.00% → a 9-point gap created by one -50% year

When a fixed sum of money buys a variable number of units, the correct average price is the harmonic mean

Prices applied to a FIXED cash amount → harmonic. Returns applied to capital → geometric. Quantities → arithmetic.

$$\bar{X}_H = n / \sum_i (1 / X_i) \quad \text{AM} \geq \text{GM} \geq \text{HM} \quad (\text{equal iff all } X_i \text{ equal})$$

CFA LM 1 EXAMPLE 6 – COST AVERAGING

An investor puts EUR 1,000 into the same stock each month for two months. Share prices: EUR 10, then EUR 15.

1. Month 1: $1,000 / 10 = 100.00$ shares. Month 2: $1,000 / 15 = 66.67$ shares.
2. Total: 166.67 shares for EUR 2,000 → average price actually paid = $2,000 / 166.67 = \text{EUR } 12.00$
3. Harmonic mean of the prices: $2 / (1/10 + 1/15) = \text{EUR } 12.00$ ✓ the arithmetic mean would say EUR 12.50

HM 12.00 < AM 12.50 – the fixed cash amount buys MORE shares when prices are low

THE SAME LOGIC PRICES A MULTIPLE

CFA Example 7 — ten stocks whose P/Es average 12.15 arithmetically have a harmonic mean P/E of 10.81 (geometric 11.43). An index P/E divides total price by total earnings, which is exactly this weighting.

¹ $\text{AM} \times \text{HM} = \text{GM}^2$ holds exactly for two observations ($12.5 \times 12.0 = 12.2474^2$). For larger samples $\text{AM} \geq \text{GM} \geq \text{HM}$ still holds — the product identity does not.

Source: CFA Institute, Vol 1, Learning Module 1, §3 "The Harmonic Mean", Eq. 4, Examples 6–7, pp. 14–17.

Trimmed and winsorized means survive an outlier that would drag the arithmetic mean anywhere it liked

Trimmed = delete the tails. Winsorized = replace them with the nearest surviving value. Both keep the centre honest.

TRIMMED MEAN

Remove a defined percentage of the largest and smallest observations, then take the arithmetic mean of what is left. The outlier contributes nothing at all.

WINSORIZED MEAN

Keep every observation but replace the extremes at both ends with the value of their nearest surviving neighbour. The outlier still votes — just quietly.

Measure	How it is built (10 annual returns)	Base case	Year 4 at -40%
Arithmetic mean	all 10 observations	3.00%	-0.80%
Geometric mean	compound all 10 gross factors	2.97%	-1.96%
Trimmed mean	drop 1 high and 1 low, average the 8	3.25%	3.25%
Winsorized mean	tails replaced by nearest kept value	3.15%	3.15%

¹ Naming: dropping one of ten from each tail is a 10% trim per tail. A "5% trimmed mean" conventionally discards the lowest and highest 2.5%.

² Base series (%): 4.5, 6.0, 1.5, -2.0, 0.0, 4.5, 3.5, 2.5, 5.5, 4.0. CFA-confirmed AM 3.00%, GM 2.9717%.

Source: CFA Institute, Vol 1, Learning Module 1, §3, pp. 17–18; return series from the LM 1 Question Set, Exhibit 9, p. 18.

Money-weighted return measures what the investor earned; time-weighted return measures what the manager did

MWR is an IRR on the investor's cash flows. TWR compounds sub-period returns, which makes cash-flow timing irrelevant.

$$\text{MWR: } \sum CF_t / (1 + \text{MWR})^t = 0 \qquad \text{TWR: } 1 + R_{\text{TW}} = (1+R_1)(1+R_2)\dots(1+R_n)$$

where: R_t = return of the sub-period between cash flows; CF_t = net investor cash flow (money in negative, money out positive). Across N years: $R_{\text{TW}} = [\prod(1+R_{\text{year}})]^{(1/N)} - 1$.

MONEY-WEIGHTED RETURN (MWR)

An IRR. Weights every sub-period by the capital actually at work, so it is the return the INVESTOR experienced given her deposits and withdrawals.

Right metric for personal wealth, and for any vehicle where the manager controls the drawdown of capital — private equity, infrastructure, real estate.

TIME-WEIGHTED RETURN (TWR)

The compound growth of one unit invested at the start, built by linking sub-period returns. Neutral to the timing and size of cash flows.

The preferred measure for evaluating PORTFOLIO MANAGERS, because additions and withdrawals are outside their control.

¹ Annualization trap: take a root only when each linked return spans a full year. CFA Example 10 — a 4-month and an 8-month return link to 21.03%, no root.

Source: CFA Institute, Vol 1, Learning Module 1, §4 "Money-Weighted and Time-Weighted Return", Eqs. 5–6, pp. 19–25.

One portfolio, two defensible answers — the gap is entirely explained by when the money was at work

Buy one share, buy a second a year later, sell both at the end of year two. Dividends of USD 5 per share per year, not reinvested.

Time	What happens	Net investor cash flow	Sub-period HPR
t = 0	Buy 1 share at USD 200	-200	—
t = 1	Dividend USD 5; buy a 2nd share at USD 225	-220	+15.00%
t = 2	Dividends USD 10; sell both shares at 235	+480	+6.67%

TIME-WEIGHTED vs MONEY-WEIGHTED, SAME DATA

Year 1 HPR = $(5 + 225 - 200) / 200 = 15.00\%$. Year 2 HPR = $(10 + 470 - 450) / 450 = 6.67\%$.

1. TWR = $[(1.15)(1.0667)]^{(1/2)} - 1 = 10.76\%$ per year
2. MWR solves $-200 - 220/(1+r) + 480/(1+r)^2 = 0 \rightarrow r = 9.39\%$ per year
3. Why MWR is lower: USD 450 was at work during the weak year (6.67%), only USD 200 during the strong one (15%)
4. Reverse the timing — big money in the good year — and MWR would sit ABOVE TWR. The direction is not a rule

TWR 10.76% grades the manager • MWR 9.39% is what the investor actually earned

Source: CFA Institute, Vol 1, Learning Module 1, "Money-Weighted Return for a Dividend-Paying Stock" and "Computing Time-Weighted Returns", pp. 20–24.

SECTION 03

Annualizing returns

Annualize by compounding the period return through a full year — and note that the compounding frequency itself changes the answer

The stated (nominal) rate is a quoting convention. The EAR is what the money actually does in a year.

$$R_{\text{annual}} = (1 + R_{\text{period}})^c - 1 \qquad \text{EAR} = (1 + r_s / m)^m - 1$$

where: c = periods per year (52 weekly, 12 monthly, 2/3 for an 18-month return); m = compounding periods per year; r_s = quoted rate.

Compounding	m	Rate per period	EAR ($r_s = 6\%$)	EAR ($r_s = 12\%$)
Annual	1	6.0000%	6.0000%	12.0000%
Semi-annual	2	3.0000%	6.0900%	12.3600%
Quarterly	4	1.5000%	6.1364%	12.5509%
Monthly	12	0.5000%	6.1678%	12.6825%
Daily	365	0.016438%	6.1831%	12.7475%
Continuous	∞	limit	6.1837%	12.7497%

¹ Annualizing assumes the period return repeats. A 5% week annualizes to 1,164.3% ($= 1.05^{52} - 1$). Use it to compare, never to forecast.

² CFA Example 13 — 4.61%/146 days = 11.93% p.a.; 1.10%/5 weeks = 12.05%; 14.35%/15 months = 11.32%. The smallest periodic return compounds most often.

Source: CFA Institute, Vol 1, Learning Module 1, §5 "Annualized Return", Eqs. 7–9 and Examples 12–13, pp. 29–32.

Continuous compounding is the limit as frequency $\rightarrow \infty$, and its returns ADD across time instead of compounding

Additivity is the whole point — it is why option models, log-return series and any non-integer horizon use this convention.

$$FV = PV \times e^{(r_{cc} \times n)} \qquad r_{cc} = \ln(1 + R) = \ln(P_1 / P_0)$$

where: $r_{cc} \leq R$ always, because $\ln(1+x) \leq x$. Over many periods $r_{(0,T)} = r_{(0,1)} + r_{(1,2)} + \dots + r_{(T-1,T)}$ — a sum, not a product.

DISCRETE $\rightarrow$ CONTINUOUS

A 12% holding-period return is a continuously compounded return of $\ln(1.12) = 11.33\%$.

A smaller number describing exactly the same amount of money.

CONTINUOUS $\rightarrow$ DISCRETE

A 10% continuously compounded rate corresponds to a discrete annual return of $e^{0.10} - 1 = 10.52\%$.

Pick one convention per model and never mix.

STRAIGHT FROM PRICES

$r_{cc} = \ln(P_1/P_0)$. Buy at USD 30, worth USD 34.50: $\ln(34.50/30) = 13.98\%$, against a 15.00% holding-period return.

No income term needed — the price relative does the work.

Source: CFA Institute, Vol 1, Learning Module 1, §5 "Continuously Compounded Returns", Eqs. 10–13, p. 32.

Gross, net, after-tax, real and leveraged returns are five progressively adjusted views of one identical asset

Assumptions: 0.85% fees, 30% tax on income and realised gains, 2.5% inflation, 50% debt costing 4%.

Measure	What is deducted, and in which order	Result
Gross return	nothing; trading commissions ARE already deducted	8.00%
Net return	– 0.85% management and administrative expenses	7.15%
After-tax nominal	– 30% tax on income and realised gains	5.01%
Real return	÷ (1 + 2.5% inflation), applied AFTER tax	2.44%
Leveraged return	50% debt at 4%: $8.00 + 1.00 \times (8.00 - 4.00)$	12.00%

$$R_L = R_P + (V_B / V_E) \times (R_P - r_D)$$

where: V_B = debt, V_E = equity, r_D = cost of debt. Leverage adds value only while $R_P > r_D$.

LEVERAGE IS SYMMETRIC

The same 50% debt that turns +8% into +12% turns –4% into –12%. Leverage never reduces risk; it redistributes outcomes. (CFA's milder case: 30% debt at 5% lifts 8% to 9.29%.)

Source: CFA Institute, Vol 1, Learning Module 1, §6 "Other Major Return Measures", Eq. 15 and Examples 14–15, pp. 33–37.

SECTION 04

Future value

The future value of a single cash flow is PV grown by $(1+r)$ over n periods — every other TVM formula is built from this one

Annuities, perpetuities, bond prices and DDMs are all just sums of this expression applied at different dates.

$$FV_n = PV \times (1 + r)^n$$

$$FV_{\text{cont}} = PV \times e^{(r \times n)}$$

where: PV = present value; r = rate PER PERIOD; n = number of PERIODS. r and n must be measured in the same unit — this is non-negotiable.

CASH-FLOW TIMELINE — ALWAYS DRAW ONE

Deposit GBP 10,000 today ($t = 0$) at 5% per year, compounded annually. What is the balance after 10 years?

1. $PV = 10,000$; $r = 5\%$; $n = 10$.
2. $(1.05)^{10} = 1.62889$
3. $FV = 10,000 \times 1.62889 = \text{GBP } 16,288.95$
4. Sense check: at 5% money doubles in about 14 years, so $\sim 1.63\times$ after 10 is right.

GBP 16,289 — a 62.9% total gain from 5.00% per year

Compounding frequency materially changes the outcome even when the nominal rate never moves

GBP 1,000 invested for 5 years at a 12% nominal rate. Annual to continuous is worth 3.4% of final value.

$$FV = PV \times (1 + r_s/m)^{(m \times n)}$$

$$FV_{\text{cont}} = PV \times e^{(r_s \times n)}$$

Compounding	m	Periods (m × n)	FV after 5 years	Uplift vs annual
Annual	1	5	GBP 1,762.34	—
Semi-annual	2	10	GBP 1,790.85	+28.51
Quarterly	4	20	GBP 1,806.11	+43.77
Monthly	12	60	GBP 1,816.70	+54.36
Daily	365	1,825	GBP 1,821.94	+59.60
Continuous	∞	—	GBP 1,822.12	+59.78 (+3.4%)

Source: CFA Institute, Vol 1, Learning Module 1, "Non-annual Compounding", Eq. 7 and Example 11, pp. 29–30; LM 2, Eq. 2, p. 46.

The future value of a level annuity is a closed-form sum of compounded payments — the core arithmetic of retirement planning

Ordinary annuity = payments at period END. Annuity due = payments at START — multiply the ordinary result by $(1 + r)$.

$$\text{FV_annuity} = \text{PMT} \times [((1 + r)^n - 1) / r]$$

where: PMT = level payment; r = rate per period; n = number of payments. The bracketed term is the future-value annuity factor.

RETIREMENT - SAVINGS EXAMPLE

Save GBP 10,000 at the end of every year for 20 years, in an account paying 6% per year.

1. $\text{PMT} = 10,000$; $r = 6\%$; $n = 20$. $(1.06)^{20} = 3.20714$
2. Annuity factor = $(3.20714 - 1) / 0.06 = 36.7856$
3. $\text{FV} = 10,000 \times 36.7856 = \text{GBP } 367,856$ of which GBP 200,000 is principal
4. Pay at the START of each year instead: $\times 1.06 = \text{GBP } 389,927$. One year of timing is worth GBP 22,071.

GBP 367,856 ordinary · GBP 389,927 annuity due · compound interest = GBP 167,856

Source: CFA Institute, Vol 1, Learning Module 2, §2 "Annuity Instruments", Eq. 8, p. 54; annuity FV form from the TVM pre-reads.

SECTION 05

Present value

Present value is future value discounted at r — and any stream of cash flows is worth the sum of its discounted parts

The discount factor $1/(1+r)^n$ is the only thing being applied anywhere in this section. Learn to read it as a price.

$$PV = FV_n / (1 + r)^n = FV_n \times (1 + r)^{-n}$$

where: Move money BACKWARDS on the timeline by dividing by the compounding factor. Continuous equivalent: $PV = FV \times e^{(-r n)}$.

Horizon	DF at 5%	PV of GBP 100
1 year	0.9524	95.24
5 years	0.7835	78.35
10 years	0.6139	61.39
20 years	0.3769	37.69
30 years	0.2314	23.14

At 5% a cash flow loses half its value every 14.2 years. At 10%, every 7.3. This single fact is why terminal value dominates almost every DCF you will build.

CASH-FLOW ADDITIVITY

Cash flows indexed to the same date can be added. So the present value of any stream is simply the sum of the present values of its parts.

That is what forces no-arbitrage: two strategies with identical dated cash flows must carry identical prices, or a riskless profit exists.

CFA Example 12 — GBP 45 / 45 / 45 and GBP 60 / 40 / 32.35 against the same GBP 100 outlay at 10% both have PV = GBP 11.91. Different shapes, identical value.

Source: CFA Institute, Vol 1, Learning Module 2, §2 Eqs. 3–5, pp. 46–48; §4 "Cash Flow Additivity", pp. 69–70.

Annuities and perpetuities have closed-form present values that collapse an entire cash-flow stream into one number

A perpetuity is an annuity with $n \rightarrow \infty$: the $(1+r)^{-n}$ term vanishes and the factor collapses to $1/r$.

PV OF AN ORDINARY ANNUITY

$$PV = PMT \times \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

GBP 10,000 a year for 20 years at 6%:

$$\begin{aligned} \text{factor} &= [1 - 1.06^{-20}] / 0.06 = 11.4699 \\ PV &= 10,000 \times 11.4699 = \text{GBP } 114,699 \end{aligned}$$

Read the factor as "years of payment you are actually buying"
— 20 payments, but only 11.5 years' worth of value.

PV OF A PERPETUITY

$$PV = PMT / r$$

A perpetual bond paying GBP 50 a year at 5%:

$$PV = 50 / 0.05 = \text{GBP } 1,000$$

Run it backwards to get the yield. CFA Example 4 — KB Financial's perpetual pays KRW 0.825 a quarter and trades at KRW 97.03, so $r = 0.825/97.03 = 0.85\%$ per quarter, or 3.40% annualized.

A discount bond is a single present-value calculation — and its price accretes to par purely by the passage of time

The US, UK and India all let investors buy individual stripped coupon or principal cash flows. That is all a discount bond is.

$$\text{PV}(\text{discount bond}) = \text{FV}_t / (1 + r)^t$$

where: No coupon, so the entire return is the pull from PV up to FV. Here annual compounding is assumed, so t is simply the number of years.

CFA LM 2 EXAMPLE 1 — 20-YEAR REPUBLIC OF INDIA PRINCIPAL STRIP

A single principal cash flow of INR 100 payable in 20 years, issued when the YTM was 6.70%.

1. $(1.067)^{20} = 3.6584 \Rightarrow \text{PV} = 100 / 3.6584 = \text{INR } 27.33$ per INR 100 of principal
2. Three years later, yields unchanged: $\text{PV} = 100 / (1.067)^{17} = \text{INR } 33.21$
3. The INR 5.88 gain is accrued interest, not a windfall — a positive r pulls PV up towards FV as t shrinks
4. Now move the yield instead: at a price of INR 22.68, $r = (100/22.68)^{(1/20)} - 1 = 7.70\%$ — a 100bp rise costs INR 4.65

INR 27.33 today · INR 33.21 in three years at constant yields · INR 22.68 if yields rise 100bp

¹ Compound the same 6.70% semiannually (40 periods at 3.35%) and the price falls to INR 26.77. The compounding convention is part of the price.

Source: CFA Institute, Vol 1, Learning Module 2, §2 Example 1 (Republic of India discount bond) and Example 3, pp. 48–52.

When the coupon equals the YTM a bond prices at par — the moment the market demands more, the price must fall

Coupon < YTM $\Rightarrow$ discount. Coupon > YTM $\Rightarrow$ premium. Coupon = YTM $\Rightarrow$ par. That is the whole of bond-pricing intuition.

CFA LM 2 EXAMPLE 2 — HELLENIC REPUBLIC 2.00% ANNUAL COUPON, 7 YEARS

Greece issued a 2.00% annual-coupon bond maturing in seven years, at an observed YTM of 2.00%.

1. $PV = \sum 2/(1.02)^t$ for $t = 1 \dots 7$, plus $100/(1.02)^7$
2. PV of the seven coupons = EUR 12.94; PV of the EUR 100 principal = EUR 87.06
3. Total = EUR 100.00 — coupon equals YTM, so the bond prices exactly at par
4. Later, with five years remaining, it traded at EUR 93.091. Solving for r gives a YTM of 3.532%

Par at issuance $\rightarrow$ EUR 93.09 once the market demands 3.53%. Price and yield move in opposite directions

Year	1	2	3	4	5	6	7
Cash flow (EUR)	2	2	2	2	2	2	102
PV at 2.00% (EUR)	1.96	1.92	1.88	1.85	1.81	1.78	88.80

Source: CFA Institute, Vol 1, Learning Module 2, §2 Example 2 (Hellenic Republic annual coupon bond), p. 51; §3 Example 9, p. 64.

A stock is the present value of its dividends — and the growth assumption you attach does most of the valuation work

Same company, same 15% required return, same GBP 1.50 dividend. Only growth changes — and the price nearly doubles.

$$P_0 = D / r$$

$$P_0 = D_0(1+g) / (r - g_L)$$

$$TV_N = D_{(N+1)} / (r - g_L)$$

where: No growth, constant growth, and the terminal value that closes a two-stage model. All three require $r > g$, or the formula returns nonsense.

NO GROWTH → **GBP 10.00**

$D = \text{GBP } 1.50$ flat, forever. $r = 15\%$.

$$P_0 = 1.50 / 0.15 = \text{GBP } 10.00.$$

A pure perpetuity — identical maths to a perpetual bond.

CONSTANT $g = 6\%$ → **GBP 17.67**

$$D_1 = 1.50 \times 1.06 = 1.59.$$

$$P_0 = 1.59 / (0.15 - 0.06) = \text{GBP } 17.67.$$

Six points of perpetual growth are worth +77% of the price. Growth in the denominator is powerful and dangerous.

TWO-STAGE $6\% \rightarrow 2\% \rightarrow$
GBP 13.05

Stage 1: D_1 1.590, D_2 1.685, D_3 1.787 →
 $PV = \text{GBP } 3.83.$

$$D_4 = 1.787 \times 1.02 = 1.823; TV_3 = 1.823 / 0.13 = 14.02; PV(TV_3) = 14.02 / 1.15^3 = \text{GBP } 9.22.$$

Total GBP 13.05 — 71% of it terminal value.

¹ Terminal-growth sensitivity after three years at 6%: $g_L = 0\% \rightarrow \text{GBP } 11.66$; $2\% \rightarrow 13.05$; $4\% \rightarrow 14.94$; $6\% \rightarrow 17.67$. Two points of g move the price 14%.

Source: CFA Institute, Vol 1, Learning Module 2, §2, Eqs. 10, 14–17, Examples 6–7 (Shipline PLC), pp. 56–59.

Run the model backwards and the market price tells you what investors are actually assuming

You never observe r and g . You observe price. A price fixes the pair (r, g) — it cannot pin down either one alone.

$$r = D_1 / P_0 + g$$

$$g = r - D_1 / P_0$$

where: Both are rearrangements of the Gordon formula: required return = dividend yield + growth. In multiples: $P_0 / E_1 = (\text{payout ratio}) / (r - g)$.

CFA LM 2 EXAMPLE 10 — COCA-COLA

Share price USD 63.00; expected dividend per share over the next year USD 1.76.

1. Dividend yield = $1.76 / 63 = 2.79\%$
2. Assume $g = 4\% \Rightarrow r = 2.79\% + 4\% = 6.79\%$
3. Assume $r = 7\% \Rightarrow g = 7\% - 2.79\% = 4.21\%$

The price fixes $r - g = 2.79\%$. Nothing more

THE SAME TEST ON A MULTIPLE

Forward P/E = payout ratio / $(r - g)$.

CFA Example 11 — a forward P/E of 28 with a 70% payout requires $r - g = 2.50\%$.

Demand a 9% return with 4.5% growth and that same payout only supports a P/E of 15.6. The market is paying 28.

So either r is far below 9%, or g is far above 4.5%, or the stock is expensive. There is no fourth option.

SECTION 06

NPV, IRR & applications

NPV states the value a project adds today — accept if positive, reject if negative, and the rule never contradicts itself

A positive NPV means the project increases firm value today by exactly that amount — not eventually, today.

$$\text{NPV} = -CF_0 + \sum CF_t / (1 + r)^t \quad \text{accept iff NPV} > 0$$

where: CF_0 = initial outlay; r = cost of capital, i.e. the required return on a project of this risk. IRR is the r that sets NPV to zero.

PROJECT EVALUATION

Initial outlay USD 1,000. After-tax cash flows of 300, 400, 400, 300 across years 1–4. Cost of capital 10%.

1. $PV(y1) = 300 / 1.10 = 272.73$ · $PV(y2) = 400 / 1.10^2 = 330.58$
2. $PV(y3) = 400 / 1.10^3 = 300.53$ · $PV(y4) = 300 / 1.10^4 = 204.90$
3. $NPV = -1,000 + 272.73 + 330.58 + 300.53 + 204.90 = +108.74$
4. Undiscounted the project returns 1,400 on 1,000 — but 291 of that "profit" is just the cost of waiting

NPV = +USD 108.74 → ACCEPT. Shareholder value rises by ~USD 109 today

Source: Discounting and IRR mechanics: CFA Institute, Vol 1, Learning Module 1, §4, Eq. 5, p. 20. The NPV decision rule itself is developed in the Corporate Issuers volume.

IRR is a useful shorthand that breaks with non-conventional cash flows, different scales, or different timing — always cross-check with NPV

IRR is the r at which $NPV = 0$. It is a good comparable and a bad decision rule.

FAILURE 1 · MULTIPLE IRRs

A cash-flow pattern that changes sign more than once (–, +, –) can satisfy $NPV = 0$ at more than one rate. There are as many candidate IRRs as sign changes.

Classic case: a mining or offshore project with a large end-of-life decommissioning outflow.

FAILURE 2 · SCALE MISMATCH

Project A: invest 100, receive 150 in one year — IRR 50%.

Project B: invest 1,000, receive 1,300 — IRR 30%.

At a 10% cost of capital, $NPV(A) = +36$ and $NPV(B) = +182$. IRR prefers A; B creates five times the value. B wins.

FAILURE 3 · TIMING MISMATCH

Front-loaded projects flatter IRR, because IRR assumes early cash is reinvested at the IRR itself.

Plot NPV against r for both projects: they cross at the crossover rate. On one side of it NPV and IRR agree, on the other they do not.

¹ IRR implicitly assumes every interim cash flow is reinvested AT the IRR — rarely true, and flattering whenever the IRR sits above the cost of capital.

² For mutually exclusive projects, NPV wins the tie every time: it is denominated in value created, and IRR is not.

Source: IRR definition: CFA Institute, Vol 1, Learning Module 1, §4, Eq. 5, p. 20. Failure modes are developed in the Corporate Issuers volume.

A level mortgage payment is the annuity formula solved for PMT — and the split between interest and principal reverses over the life of the loan

The outstanding balance at any date is simply the present value of the payments still to come.

$$A = r \times PV / [1 - (1 + r)^{-t}]$$

where: PV = loan principal; r = rate PER PERIOD; t = number of payments. For a monthly loan, r is the annual rate divided by 12 and t is months.

CFA LM 2 EXAMPLE 5 — USD 800,000 MORTGAGE • 30 YEARS • 5.25% NOMINAL

80% of a USD 1,000,000 property. Monthly rate = 5.25% / 12 = 0.4375%; t = 360 payments.

1. Denominator: $1 - (1.004375)^{-360} = 1 - 0.20772 = 0.79228$
2. $A = 0.004375 \times 800,000 / 0.79228 = 3,500 / 0.79228 = \text{USD } 4,417.63$ per month
3. Month 1: interest 3,500.00, principal 917.63. Month 360: interest 19.24, principal 4,398.39
4. Total paid = $4,417.63 \times 360 = \text{USD } 1,590,347$, of which USD 790,347 is interest

USD 4,417.63 a month • interest is 79% of the FIRST payment and 0.4% of the last

Source: CFA Institute, Vol 1, Learning Module 2, §2 "Annuity Instruments", Eq. 8 and Example 5 (Mortgage Cash Flows), pp. 54–55.

Every TVM problem reduces to one question: where on the timeline is each cash flow, and at what rate do we move it?

These six archetypes cover every calculation in this deck. Identify the pattern first; reach for the formula second.

Archetype	Formula	Where you meet it
Single CF — FV	$FV = PV \times (1 + r)^n$	Savings, accretion
Single CF — PV	$PV = FV / (1 + r)^n$	Zero-coupon, strip
Ordinary annuity — PV	$PV = PMT \times [1 - (1+r)^{-n}] / r$	Mortgage, term loan
Ordinary annuity — FV	$FV = PMT \times [(1+r)^n - 1] / r$	Pension, sinking fund
Perpetuity — PV	$PV = PMT / r$	Perpetual, preferred
Growing perpetuity — PV	$PV = PMT / (r - g)$	Gordon DDM, terminal

FOUR-STEP DECISION RULE

1. Draw the timeline. 2. Identify the archetype — or decompose the stream into archetypes, which cash-flow additivity always allows. 3. Apply the matching formula. 4. Check units: does r cover exactly one period of n ?

Step 2 is the one that matters. Any cash-flow stream can be split into pieces you already know how to price, and the pieces can simply be added.

Source: CFA Institute, Vol 1, Learning Modules 1–2, summaries; cash-flow additivity, LM 2 §4, p. 69.

Deck 1 — Time Value of Money & DCF Foundations

■ r is a required return, a discount rate and an opportunity cost at once — and decomposes into real risk-free + inflation + default + liquidity + maturity.

■ Time-weighted return grades the manager; money-weighted (an IRR) is what the investor experienced. USD 200/220/480 example: 10.76% vs 9.39%.

■ Annualize by compounding: $R = (1 + R_{\text{period}})^c - 1$. $EAR = (1 + r_s/m)^m - 1$. Continuous is the limit, and continuous returns add across time.

■ Bond price = PV of coupons + PV of principal. India 20y strip at 6.70% → INR 27.33. Greece 2% at 2.00% YTM → par; at 3.532% → EUR 93.09.

■ $AM \geq GM \geq HM$. Arithmetic for next period, geometric for what capital actually did, harmonic when a fixed sum buys a variable quantity.

■ Trimmed and winsorized means barely move when one observation goes to -40%; the arithmetic mean collapses from +3.00% to -0.80%.

■ Know which of gross / net / after-tax / real / leveraged you are being shown: one 8% asset can be quoted anywhere from 2.44% to 12.00%.

■ DDM: Shipline is GBP 10.00 flat, 17.67 at 6% growth, 13.05 two-stage — 71% terminal. Invert it and the price hands you the market-implied $r - g$.